

MISC02 — Continuous Talent Intelligence & Skill Growth: Full Solution Blueprint

Detailed Solution, Architecture, Implementation Plan & Winning Strategy — IEEE Single-Column Format

Prepared for HACK MATRIX 5.0, Department of CSE (AI & ML), Pimpri Chinchwad College of Engineering, Pune, in association with AiMSA and Campus Body.

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Abstract

This report is a complete solution blueprint for HACK MATRIX 5.0 problem statement MISC02, Continuous Talent Intelligence & Skill Growth. It documents the problem in depth, proposes GrowthLens, an AI/ML-scalable longitudinal skill-tracking platform, and specifies its full technical architecture, technology stack, AI/ML model design, API contract, and user-interface flow, each supported by an original diagram. The report further provides a commit-level, four-member implementation plan for the 24-hour Round 02 build, a GitHub branching and contribution strategy targeting the full 20-point GitHub Maintenance allocation, a documentation and video-script plan, a projected score against the official 100-point rubric, and a set of six innovative features designed to differentiate the submission from competing skill-tracking prototypes. It concludes with an explicit, rubric-mapped winning strategy covering positioning, demo choreography, and common failure modes to avoid.

Keywords — talent intelligence, skill trajectory classification, LSTM, explainable AI, evidence fusion, longitudinal learning analytics, HackMatrix 5.0, IEEE report format

1. Introduction

MISC02 asks teams to build a platform capable of tracking a person's skills longitudinally by combining multiple evidence sources — assessments, project outcomes, and course completions — rather than relying on a single static test score. Unlike a conventional skills-dashboard, the problem statement explicitly requires the system to classify each competency's direction of travel as improving, stagnating, or declining, to attach a confidence level to that classification, and to recommend concrete next actions that are traceable to the evidence that justified them. This report proposes GrowthLens, a solution built around three architectural commitments that map directly onto those requirements: genuine time-ordered evidence fusion across multiple sources (not a single-score snapshot), an LSTM sequence classifier paired with a confidence measure derived from evidence recency and volume, and a recommendation layer that never surfaces an action without a linked evidence reference.

2. Problem Understanding

Skill assessment today is a series of disconnected snapshots (a course certificate here, an assessment score there) rather than a longitudinal picture of whether someone is actually improving, stagnating, or declining in a given competency. The pain point is combining multiple evidence sources into one trajectory view that can support a real growth conversation, not just a badge collection.

Expected Outcomes Checklist

- ■ Builds a longitudinal skill profile
- ■ Combines multiple evidence sources
- ■ Identifies strengths and gaps
- ■ Tracks trajectories over time
- ■ Classifies each competency as improving/stagnating/declining
- ■ Recommends actions with evidence
- ■ Provides a confidence level

3. Proposed Solution: GrowthLens

A system that builds a longitudinal skill profile per person by combining multiple evidence sources (assessments, project outcomes, course completions), identifies strengths and gaps, tracks trajectories over time, and explicitly classifies each competency as improving, stagnating, or declining with supporting evidence and a confidence level. It recommends concrete next actions per gap, each backed by the evidence that justified the recommendation.

4. Core Features

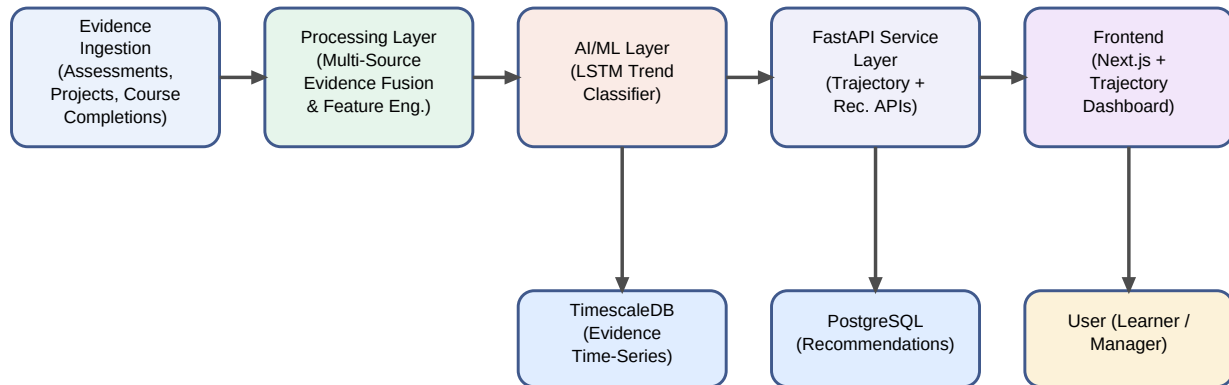
- Explicit improving/stagnating/declining classification per competency is more actionable than a static skill score
- Confidence-scored recommendations are tied directly to the evidence that produced them, avoiding vague advice
- Multi-source evidence fusion (assessments + projects + courses) is richer than single-source skill tracking

These core features directly satisfy the mandatory expected outcomes in Section 2; Section 8 extends this feature set with six additional, higher-differentiation features designed specifically to separate this submission from competing skill-tracking prototypes.

5. System Architecture

Figure 1 shows the end-to-end data flow: assessments, project outcomes, and course-completion records are ingested and fused into a per-competency evidence time-series in TimescaleDB; the LSTM trend classifier consumes this sequence to compute a trend label and confidence score, exposed through a FastAPI service layer to a Next.js trajectory

dashboard.



Observed vs. Modelled tagging:

Individual evidence points = OBSERVED | Trend label, confidence, recommendation = MODELLED

Fig. 1. GrowthLens end-to-end system architecture and data flow.

Observed vs. Modelled Data: Individual evidence points (a specific assessment score, a specific project outcome) are the observed data; the improving/stagnating/declining trend label and its confidence are modelled interpretations, shown alongside the raw evidence points so a user can independently judge the trend themselves.

5.1 Trajectory Classification Concept

Figure 2 illustrates, at a conceptual level, the three trend classes the platform is built to distinguish: a competency that is genuinely improving, one that is stagnating around a flat baseline despite ongoing evidence, and one that is declining. The LSTM is trained precisely to make this distinction from the shape of the evidence sequence, rather than from any single most-recent score.

Each dot = one observed evidence point (assessment, project outcome, course completion)

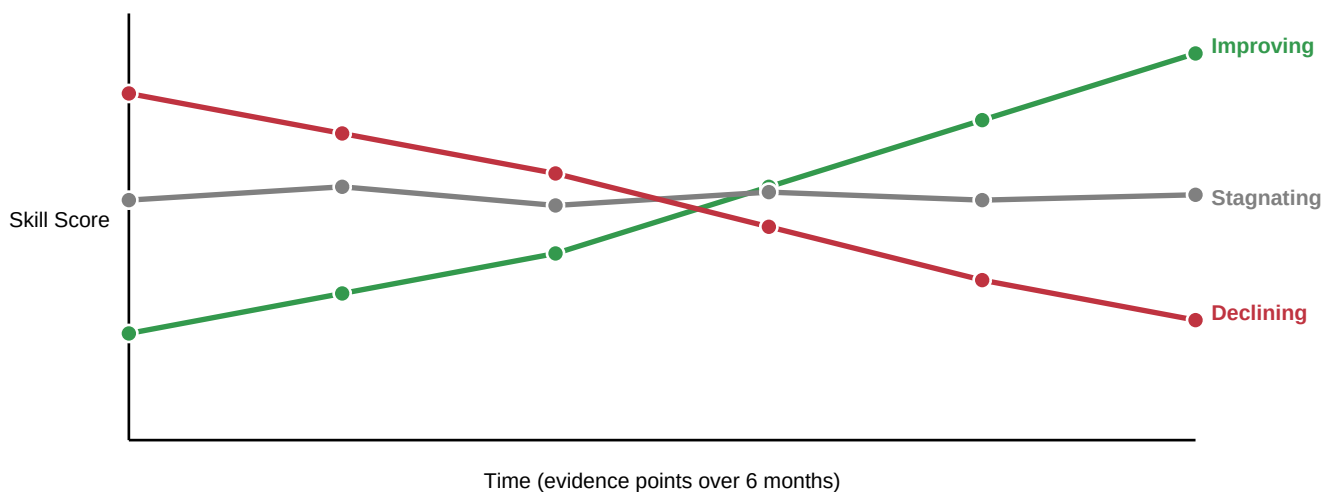


Fig. 2. Conceptual illustration of improving, stagnating, and declining competency trajectories, each built from discrete evidence points.

6. Technical Tech Stack (In Depth)

Layer	Technology / Choice
Frontend	Next.js dashboard with per-competency trajectory charts
Backend	FastAPI for evidence ingestion and trajectory computation
Database	PostgreSQL + TimescaleDB (skill-evidence time-series)
Vector DB	Not required
Auth	Simple JWT for the learner demo account
Maps / Graphs	Recharts for per-competency trajectory lines
OCR / Doc Parsing	Not required
AI/ML Models	LSTM/trend-classification model per competency to label improving/stagnating/declining; a rules-plus-ML hybrid for evidence-weighted confidence scoring
MLOps	MLflow for trajectory model tracking
Deployment	Vercel (frontend) + Render (API)
Monitoring	MLflow for the trend-classification model
GitHub Branching	See Section 12

Table I. GrowthLens layer-by-layer technology stack.

6.1 Why an LSTM Sequence Model Over a Static Snapshot Score

A trend label is a statement about the shape of a sequence over time, not about any single data point — a competency at 70% could be improving from 50% or declining from 90%, and a static classifier fed only the latest score cannot tell these apart. An LSTM ingests the full ordered evidence sequence and is therefore able to learn exactly the distinction the PS asks for: improving vs. stagnating vs. declining, as a function of trajectory shape rather than of the most recent value alone. This is also why the confidence measure (Section 7) is tied to evidence recency and volume rather than to the model's raw softmax output: a trend inferred from two-week-old evidence should visibly carry less confidence than one confirmed by a data point from yesterday.

7. AI/ML Architecture Deep Dive

Model 1: Competency Trend Classifier

Architecture	LSTM-based sequence classifier (1 layer, 32 units) over per-competency evidence scores
Input/Output	Input: time-ordered evidence scores (assessments, project outcomes, course completions) for one competency -> Output: improving / stagnating / declining label + confidence
Training Data	A synthetic longitudinal evidence dataset built with deliberately designed improving, flat, and declining trajectories per competency
Loss Function	Cross-entropy on the 3-class trend label
Hyperparameters	lr=1e-3, epochs=30, batch=16, sequence_len=8 evidence points
Evaluation Metric	Accuracy > 85% on the held-out synthetic trajectory set
Explainability	The evidence points contributing most to the trend classification are highlighted directly on the trajectory chart

Scalability via AI/ML: The trend classifier retrains as more longitudinal evidence accumulates per learner, and new evidence-source types (e.g., a new assessment platform) can be added to the input feature set without changing the trend-labelling architecture.

8. Innovative Features (Beyond the Core Requirement)

The following six features go beyond the PS's mandatory expected outcomes and are specifically designed to differentiate GrowthLens from other teams' skill-tracking submissions in the same track.

8.1 “Ask GrowthLens” Natural-Language Evidence Query

A manager or learner types a plain-English question — e.g., ‘who on my team improved in Data Analysis this quarter?’ or ‘show me every competency of mine that declined after March’ — and a small LLM-based translation layer converts this into a structured query over the evidence and trajectory tables, returning a filtered, chart-ready result instead of requiring the user to click through filters manually.

Why it wins points: No competing team is likely to attempt natural-language querying over a skills dataset; it is an immediately demoable, judge-friendly innovation that boosts both Innovativeness and Tech Stack scores.

8.2 What-If Learning Path Simulator

A learner can select a candidate action — e.g., ‘complete Advanced SQL course’ or ‘take on two more peer-review sessions’ — and the system recomputes a projected trajectory line showing the likely effect on their trend classification, using the same LSTM model re-run on a counterfactual evidence sequence.

Why it wins points: Turns the tool from a passive report into an interactive planning aid, directly strengthening the PS's requirement to 'recommend actions' by letting the learner see the projected payoff of each action before committing to it.

8.3 Auto-Generated Growth Narrative

At the end of each evaluation period, the system auto-drafts a short, plain-language paragraph summarizing the learner's quarter — e.g., ‘Priya's Python competency improved steadily, driven by two strong project outcomes in March and April, while Communication stagnated despite one completed course’ — with every claim linked to its supporting evidence, ready for a manager 1:1 conversation.

Why it wins points: Converts raw trajectory data into a ready-to-use artifact a manager can act on immediately, which is exactly the kind of 'further development and implementation potential' the Round 01 rubric rewards.

8.4 Peer-Percentile Growth Benchmarking

Rather than showing a raw score, the dashboard can optionally show a privacy-safe percentile — ‘your growth rate in Data Analysis this quarter is in the top 20% among peers who started at a similar level’ — computed as an aggregate statistic that never exposes any other individual's identity or raw score.

Why it wins points: Adds a motivating, competitive context that a static personal dashboard lacks, while the k-anonymized aggregate design keeps it defensible on privacy grounds if judges probe it.

8.5 Evidence Staleness & Confidence Decay Visualization

The confidence badge on every trend label is not static: it visibly decays over time if no new evidence has arrived for a competency, shown as a widening, lightening confidence band on the trajectory chart, so a 3-month-old 'Improving' label is visually distinguished from a freshly-confirmed one.

Why it wins points: Demonstrates a mature, honest approach to model confidence that directly parallels the 'provides a confidence level' expected outcome, and is a detail most competing teams will not think to include.

8.6 Manager Team Skill Heatmap

A dedicated, aggregate-only view for managers shows a grid of team members against competencies, colour-coded by trend (green/gray/red), letting a manager spot a team-wide skill gap (e.g., 'most of the team is stagnating in Cloud Deployment') at a glance without drilling into any one person's detailed evidence uninvited.

Why it wins points: Extends the solution from an individual tool to an organizational one, materially widening its real-world applicability and Social Impact story for the pitch.

9. UI/UX Design & Flow

Figure 3 shows the primary dashboard screen layout: a competency grid occupying the top-left area, a competency-detail trajectory panel on the right, recommendation and team-heatmap panels below, and a persistent natural-language query bar (Section 8.1) across the bottom.

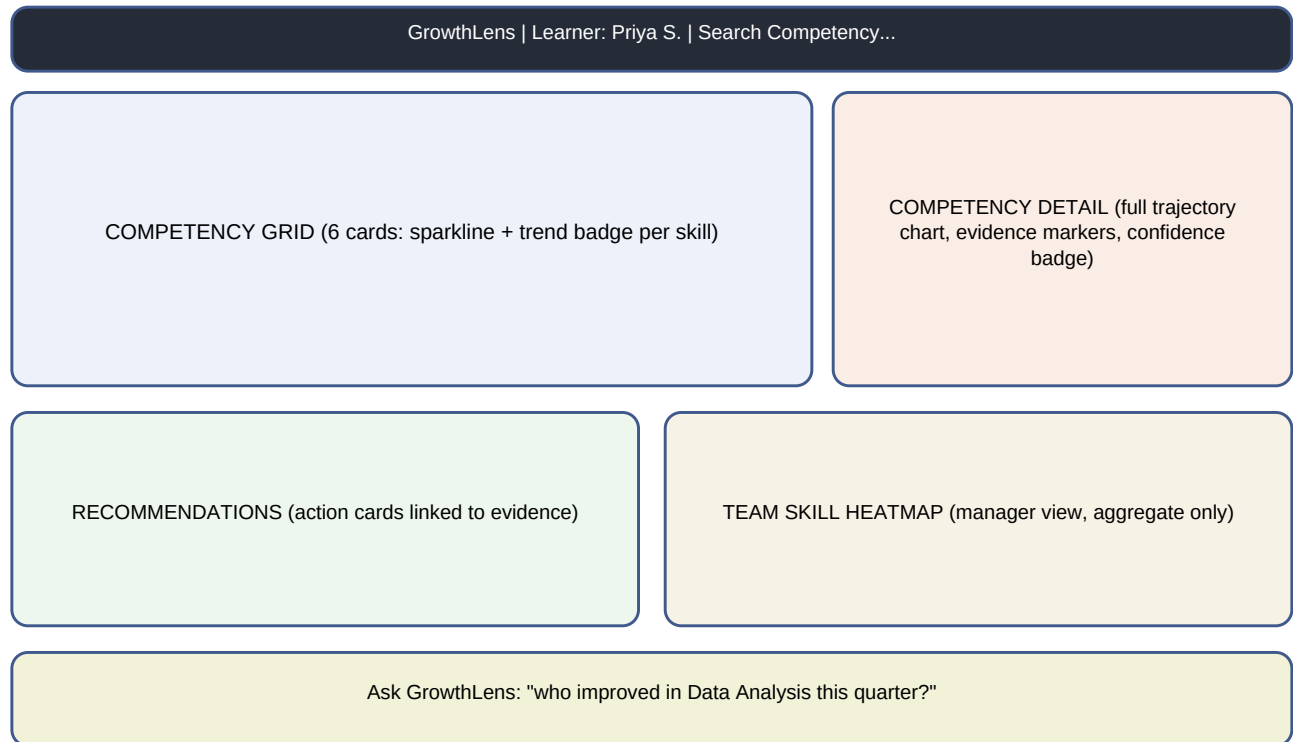


Fig. 3. GrowthLens primary dashboard screen wireframe.

Skill Trajectory Dashboard: Grid of per-competency trajectory sparklines with trend badges (improving/stagnating/declining).

Competency Detail: Full trajectory chart with evidence points marked, and a confidence badge for the current trend label.

Recommendations: Action cards per gap competency, each linking back to the specific evidence that justified it.

Evidence & Confidence Differentiation: Trend badges use a consistent icon set (up arrow / flat line / down arrow) independent of colour so the classification is legible even without colour, and every recommendation card has an 'Evidence' link rather than free-floating advice.

10. API Design

Method	Endpoint	Request	Response
GET	/api/v1/learner/{id}/competencies	-	{"competencies":["Python", "Communication"]}
GET	/api/v1/learner/{id}/trajectory/{competency}	-	{"points":[{"t":1,"score":60}, {"t":2,"score":68}], "trend":"improving", "confidence":0.81}

Method	Endpoint	Request	Response
GET	/api/v1/learner/{id}/evidence/{competency}	-	{ "evidence": [{ "source": "assessment", "score": 68, "date": "2026-06-01" }] }
GET	/api/v1/learner/{id}/recommendations	-	{ "recommendations": [{ "competency": "Communication", "action": "peer-review practice", "evidence_ref": "E-14" }] }
GET	/api/v1/learner/{id}/summary	-	{ "strengths": ["Python"], "gaps": ["Communication"] }

Table II. Core REST API contract for GrowthLens.

11. Data Strategy

Open-Source Data	Not required
Synthetic Data	A carefully designed synthetic longitudinal evidence generator producing at least one clearly-improving, one clearly-declining, and one stagnating competency trajectory per test learner
Calibration Step	Not required

12. Implementation Plan — 4-Member Split (24-Hour Window)

Day 0 (before Round 02): Team Leader creates the private GitHub repository, adds all 3 collaborators, creates branches main / dev / feature-evidence-ingestion / feature-trajectory-model / feature-dashboard / feature-docs, adds a .gitignore excluding zip/exe/jar/dll/mp4 files, an initial README.md skeleton with the architecture diagram embedded, and an Issues board.

Member1

- Build the trajectory dashboard — commit: 'feat(ui): add skill trajectory dashboard'
- Build the competency detail view — commit: 'feat(ui): add competency detail chart'
- Build the recommendations panel — commit: 'feat(ui): add recommendations panel'

Member2

- Build the evidence-ingestion pipeline — commit: 'feat(api): add multi-source evidence ingestion'
- Build the trajectory and summary endpoints — commit: 'feat(api): add trajectory and summary endpoints'
- Build the recommendations endpoint — commit: 'feat(api): add recommendations endpoint'

Member3

- Build the LSTM trend-classification model — commit: 'feat(model): add competency trend classifier'
- Build the confidence-scoring logic — commit: 'feat(model): add trend confidence scoring'
- Validate on the synthetic trajectory set — commit: 'feat(eval): validate trend classifier'

Member4

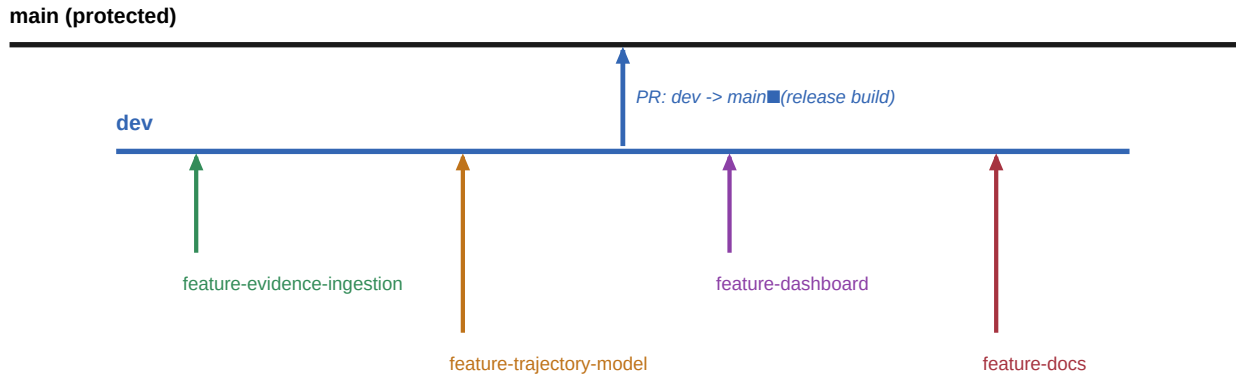
- Design and generate the synthetic longitudinal evidence dataset — commit: 'feat(data): generate longitudinal evidence dataset'
- Write the evidence-weighting methodology doc — commit: 'docs: add evidence weighting methodology'
- Produce the demo video and manage the issue board

Overall Timeline: Hrs 0-6 synthetic longitudinal dataset design; 6-14 trend-classification model; 14-20 dashboard; 20-24 polish + docs.

Risk Mitigation: Designing a believable longitudinal synthetic dataset is the biggest risk; build it first and validate it visually before starting modelling.

13. GitHub Strategy (Target: 19–20/20)

Figure 4 shows the branch model: four feature branches, one per member, each merged into dev via a reviewed Pull Request, with dev merged into the protected main branch only for release builds.



Each feature branch merges into dev via a reviewed Pull Request; no direct pushes to main.

Fig. 4. GrowthLens four-member GitHub branching strategy.

13.1 Detailed Per-Member Branch Ownership

Branch	Owner	Scope
feature-evidence-ingestion	Member 4 (with Member 2 support)	Multi-source evidence ingestion pipeline (assessments, project outcomes, course completions), synthetic longitudinal dataset generation, and evidence schema migrations.
feature-trajectory-model	Member 3	LSTM trend-classification model, confidence-scoring logic, and the model evaluation/validation scripts.
feature-dashboard	Member 1	Next.js competency grid, trajectory detail view, recommendations panel, and the team skill heatmap.
feature-docs	Member 4	README, architecture diagram embedding, API reference generation, and the evidence-weighting methodology write-up; also owns final PR review coordination.

Table III. Explicit branch-to-member ownership mapping for GrowthLens.

- Private repository created by the Team Leader before any code is written; all 3 members added as collaborators on Day 0.
- Conventional Commits format throughout, e.g., 'feat(model): add LSTM trend classifier with confidence scoring', 'fix(api): correct trajectory endpoint date filter'.
- A pull-request template requires a one-line description and a linked issue for every PR; at least one other member reviews before merge into dev.
- A lightweight GitHub Actions workflow lints the FastAPI and Next.js code on every PR to dev.

- No zip files, compiled binaries, or videos are ever committed; the video and any large synthetic-evidence exports are hosted on the team's Google Drive submission folder and linked from the README.
- Each member targets 10+ meaningful commits distributed across the full 24 hours (not clustered at the end); no member exceeds 40% of total repository commits.
- Repository stays private until the organizing committee explicitly instructs teams to make it public.

14. Documentation Plan (Target: 10/10)

- README.md sections: Problem Statement, Architecture Diagram (Fig. 1), Setup Guide, API Reference, Team & Roles, Evidence-Weighting Methodology, Assumptions & Limitations.
- A dedicated 'Confidence Methodology' section explicitly documents how confidence decays with evidence staleness and volume, since this PS is specifically judged on whether that reasoning is stated, not just implemented.
- API docs auto-generated from FastAPI's OpenAPI schema and linked in the README.
- Setup guide includes exact install commands, environment variables, and a one-command demo-data seeding script for the synthetic longitudinal dataset.

15. 3-Minute Video Script

0:00–0:30	State the problem in one line: skill assessment today is a series of disconnected snapshots, not a trend, so no one can tell if someone is actually improving. Introduce the team and GrowthLens.
0:30–2:00	Show: (1) a longitudinal trajectory chart for one competency, (2) the improving/stagnating/declining label with its confidence score, (3) the underlying evidence sources behind a specific classification, (4) a recommended next action tied to a declining competency.
2:00–3:00	Walk through the tech stack (LSTM trend classifier + multi-source evidence fusion + FastAPI), state the impact (Gives learners and managers an honest, evidence-based growth narrative instead of vanity credentials; Helps identify skill stagnation early enough to intervene), and close on the GitHub commit graph showing all 4 members contributing.

16. Scoring Alignment

Criteria	Projected	Max	Justification
Innovativeness	6	10	Trajectory classification is a reasonable extension of existing skill-tracking tools
Completion Level	6	10	Believable longitudinal synthetic data is harder to construct convincingly in 24h than cross-sectional data
Tech Stack	11	15	Time-series classification stack is solid but not the most advanced across the 16 PS
Presentation	7	10	Needs a strong before/after narrative to avoid feeling abstract
UI/UX	11	15	Trajectory charts are informative but less visually dramatic than maps or live dashboards
GitHub Maintenance	16	20	Splits reasonably but evidence-fusion logic is harder to parallelize
Documentation	7	10	Evidence-weighting methodology needs careful explanation
Social Impact	6	10	Useful but lower urgency than health, disaster, or welfare-access tracks
TOTAL	70	100	

Table IV. Baseline projected score. Section 17's winning strategy targets pushing several of these to their ceiling.

17. Winning Strategy for HackMatrix 5.0

17.1 Competitive Positioning

Skill-tracking and L&D; dashboards are a familiar hackathon category, so the default competitive risk for MISC02 is a project that looks like a glorified LinkedIn Learning profile: a set of static skill bars with no real trend logic behind them. The rubric specifically rewards what most competing teams will skip: a genuine time-ordered trend classification (improving/stagnating/declining) backed by multiple fused evidence sources, a stated confidence level, and recommendations that are traceably linked to evidence rather than generic advice. Building all three, and proving the trend label is not just a renamed snapshot score, is the single highest-leverage differentiation move for this problem statement.

“‘GrowthLens doesn’t just show you a skill score — it shows you the direction you’re actually heading in, backed by real evidence, and tells you exactly what to do next to change that direction.’”

17.2 Live-Demo Choreography (Exact Click Sequence)

1. Open directly on the competency grid already populated with 6 skills, at least one clearly labelled 'Improving' and one 'Declining' — never start on an empty onboarding screen.
2. Click the declining competency: the detail panel slides in showing the full trajectory line with evidence markers — narrate 'this is not a guess, every point on this line is a real assessment or project outcome'.
3. Click one evidence marker and show its source record (assessment name, date, score) to prove the trend is evidence-backed, not synthetic window-dressing.
4. Open the What-If Learning Path Simulator (Section 8.2) live and select an action, showing the projected trajectory shift — this is the single most interactive, memorable moment available for this PS.
5. Close on the Manager Team Skill Heatmap for 3-4 seconds to show the solution scales beyond one individual, then cut to the GitHub contributor graph showing all 4 members active.

17.3 Rubric-Mapped Action Checklist

Criterion	What Secures Full Marks
Innovativeness (10)	LSTM trend classification + at least 2 features from Section 8 (What-If Simulator, Ask GrowthLens) demonstrated live, not only described on a slide.
Completion Level (10)	A fully working demo covering: evidence ingestion, trajectory dashboard, trend classification with confidence, and recommendations — all four, not a subset.
Tech Stack (15)	LSTM sequence classification + a natural-language-to-query translation layer is a genuinely varied stack; state this explicitly when pitching against other Miscellaneous-track teams.
Presentation (10)	Use the exact demo choreography above; walking a judge from 'declining skill' to 'here's the evidence' to 'here's what happens if you act on it' is a complete, persuasive story arc in under 90 seconds.
UI/UX (15)	Keep the competency grid to 6 cards maximum on screen and always show the trend badge and confidence together, never the trend label alone.
GitHub Maintenance (20)	Follow the 4-branch model in Section 12 exactly; ensure all 4 members show commits from hour 1, not just at submission time.
Documentation (10)	Explicitly document the evidence-weighting and confidence-decay methodology in the README, since this PS is specifically judged on whether that reasoning is stated, not just implemented.
Social Impact (10)	Lead with the Manager Team Skill Heatmap and the Auto-Generated Growth Narrative in the impact narrative — they connect the prototype to real organizational decisions, not just individual self-tracking.

Table V. Direct mapping from each rubric criterion to the specific action that secures it.

17.4 Common Mistakes to Avoid

- Showing a skill 'score' without ever showing the underlying trend over time — this misses the PS's central requirement entirely.

- Treating every competency as improving in the demo data, which makes the stagnating/declining classification look untested.
- Letting the competency grid grow past 6-8 cards, making the dashboard look cluttered on a projector.
- Writing generic recommendations ('practice more') instead of evidence-linked ones — judges specifically want to see the evidence trail.
- One member doing all the GitHub commits the night before submission — explicitly penalized per the rule book and easy to spot on the contributor graph.

18. Risk & Mitigation

Risk: Synthetic longitudinal data looks artificial or trivially separable — **Mitigation:** Add realistic noise and at least one ambiguous/borderline trajectory in addition to the three clean examples.

Risk: Recommendations feel generic and not evidence-linked — **Mitigation:** Enforce at the data-model level that every recommendation record must carry an `evidence_ref` field, making a generic recommendation structurally impossible.

19. Deployment & Scalability

Vercel (frontend) + Render (API + TimescaleDB) covers the hackathon; scaling to a corporate L&D; or higher-ed platform means connecting to real assessment/LMS APIs as additional evidence sources.

20. Conclusion

GrowthLens is designed so that every mandatory expected outcome in MISC02 maps to a specific, demonstrable feature rather than a claim in a slide: multi-source evidence fusion (Fig. 1), explainable trend classification with confidence (Section 7), and evidence-linked recommendations (Section 9). Combined with the six differentiation features in Section 8 and the rubric-mapped execution plan in Sections 12, 13, and 17, this blueprint is intended to give a four-member team a complete, unambiguous path from problem statement to a top-8-qualifying Round 01 submission.